

Energy Sheet

Name _____

1 · Colour the Hundred

These hundred squares are all the energy Japan uses in a year. Colour **16** squares one colour — that is everything Japan makes for itself. Colour the other **84** a different colour. All of those came on a ship.

**16 squares**

Made in Japan

**84 squares**

Arrived by ship

When you have finished, stand back and look at it from across the room.

Then write one sentence here about what you see:

2 · Power Words

Japanese builds big words out of small pictures. 力 means *power* — it looks like an arm pushing. Glue it onto another picture and you get a kind of power. Trace the two grey copies in each box.

火力

FIRE POWER

coal and gas

火力 火力

水力

WATER POWER

rivers

水力 水力

風力

WIND POWER

windmills

風力 風力

電気

ELECTRICITY

lightning + air

電気 電気

3 · My Power Island — the checklist

Draw your island on a big sheet of plain paper. Tick a box when your drawing really has that thing on it. All four, or the lights go out.

☐
Something that works in the **daytime**.
☐
Something that works at **night**.
☐
Something for a **still, cloudy day**.
☐
A place to **keep energy for later**.

Everything else in this lesson happens on a screen and on plain paper. The six stations are at wonderschool.vercel.app → 2nd grade → Energy. Do your drawings — the chain from your lamp, your own power picture-word, and your island — as big as you like on blank paper.

Power Source Cards

✂ Cut along the borders · nine cards
Rules and a sortable table: wonderschool.vercel.app

BURNING BLACK ROCK

COAL

Runs	75%
Carbon	820 g
Cost	2 / 5
Japanese	0 / 3
At night	3 / 3
Land	2 / 5

Over a quarter of Japan's electricity. The dirtiest card in the pack, and not one gram of it comes from Japan.

CHILLED TO -162 °C

GAS · LNG

Runs	55%
Carbon	490 g
Cost	3 / 5
Japanese	0 / 3
At night	3 / 3
Land	1 / 5

Japan started this trade. First tanker into Tokyo Bay, 1969. Half the carbon of coal, and it can speed up fast.

THE EMERGENCY CARD

OIL

Runs	35%
Carbon	650 g
Cost	5 / 5
Japanese	0 / 3
At night	3 / 3
Land	1 / 5

Only 2% of the electricity — but nearly every car and plane. 93% of it comes through a gap 21 miles wide.

SPLITTING ATOMS

NUCLEAR

Runs	80%
Carbon	12 g
Cost	2 / 5
Japanese	3 / 3
At night	3 / 3
Land	2 / 5

54 reactors before March 2011. Every one switched off afterwards. Fifteen have come back.

PANELS ON ROOFS

SOLAR

Runs	15%
Carbon	48 g
Cost	2 / 5
Japanese	3 / 3
At night	0 / 3
Land	5 / 5

About 103 gigawatts, sixth in the world. Free fuel, no ship — and absolutely nothing after sunset.

BLADES IN THE SKY

WIND

Runs	25%
Carbon	11 g
Cost	3 / 5
Japanese	3 / 3
At night	1 / 3
Land	4 / 5

The cleanest card here. Japan's sea floor drops away fast, so the turbines have to float — which is dear.

FALLING WATER

HYDRO

Runs	45%
Carbon	24 g
Cost	1 / 5
Japanese	3 / 3
At night	3 / 3
Land	4 / 5

Japan's oldest electricity and still its cheapest. Nearly every good river already has a dam on it.

STEAM FROM HOT ROCKS

GEOTHERMAL

Runs	70%
Carbon	38 g
Cost	4 / 5
Japanese	3 / 3
At night	3 / 3
Land	2 / 5

Japan sits on volcanoes and takes under half a percent from them. The best spots are hot-spring towns.

CHIPS AND RUBBISH

WOOD & WASTE

Runs	65%
Carbon	230 g
Cost	5 / 5
Japanese	1 / 3
At night	3 / 3
Land	3 / 5

Burning the town's rubbish beats burying it. The wood pellets, though, are shipped in from Canada and Vietnam.

Runs = how much of the year it really produces · **Carbon** = grams of CO₂ per kWh over the machine's whole life ([IPCC AR5, Annex III](#); the oil figure is an estimate) · **Japanese** = how much Japan can supply itself · **Cost** and **Land** are simplified game scores. · Before you call a category, say out loud why it matters.